

✓ 1. Where Browser Data Is Stored (Path + Artifacts)

All browser artifacts are SQLite databases stored in the user's profile folder.

🔍 Google Chrome Artifacts

Path:

C:\Users\<USER>\AppData\Local\Google\Chrome\User Data\Default\

✓ Files to Analyze:

Artifact	File Name
Browsing History	History (SQLite DB)
Downloads	History (downloads table)
Bookmarks	Bookmarks JSON
Cookies	Cookies (SQLite DB)
Saved Passwords	Login Data
Autofill	Web Data
Extensions	Extensions\ folder
Cache	Cache\ folder

🔍 Microsoft Edge Artifacts

Path:

C:\Users\<USER>\AppData\Local\Microsoft\Edge\User Data\Default\

Same files as Chrome:

- History
- Login Data
- Web Data
- Cookies
- Extensions\

🔍 Mozilla Firefox Artifacts

Path:

C:\Users\<USER>\AppData\Roaming\Mozilla\Firefox\Profiles\<profile>.default\

✓ Files:

Artifact	File Name
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Browsing History	places.sqlite
Downloads	downloads.sqlite
Saved Passwords	logins.json
Cookies	cookies.sqlite
Cache	cache2\
Extensions	extensions.json

Brave Browser Artifacts

Path:

C:\Users\<USER>\AppData\Local\BraveSoftware\Brave-Browser\User Data\Default\
Files are exactly like Chrome:

- History
- Login Data
- Cookies
- Web Data
- Extensions\

2. TOOLS TO ANALYZE BROWSER ARTIFACTS

These tools automatically parse browser databases:

1. Autopsy (Free)

Steps:

- Load forensic image → Add data source
- Modules → **Recent Activity**
- Autopsy extracts:
 - Browsing history
 - Cookies
 - Downloads
 - Search terms
 - Extensions
 - Cache
 - Login data (encrypted unless Chrome logged in)

2. Browser History Examiner / Browser Forensics Tool

Parses:

- Chrome
- Firefox
- Edge

- Brave

Extracts:

- History
- Downloads
- Cookies
- Search keywords
- URLs visited
- Deleted history (if recoverable)

3. NirSoft Tools

Very useful for Windows systems:

✓ Browsing History

BrowserHistoryView.exe

✓ Downloads

ChromeDownloadsView.exe

✓ Cookies

ChromeCookiesView.exe

✓ Passwords (Local Profile Only)

⚠ Works only if Windows password is known

ChromePass / WebBrowserPassView

4. Magnet AXIOM / EnCase / FTK

Professional tools showing:

- Full browsing timeline
- Web cache
- Session tokens
- Web form autofill
- Cloud sync data
- Deleted history recovery

✓ 3. HOW TO EXTRACT EACH ARTIFACT

🔥 A. Browsing History

Chrome/Edge/Brave file:

History

Query it manually:

```
SELECT url, title, visit_count, last_visit_time  
FROM urls
```

ORDER BY last_visit_time DESC;

Firefox file:

places.sqlite

Query:

```
SELECT url, title, last_visit_date  
FROM moz_places;
```

B. Download History

Chrome/Edge/Brave:

```
SELECT target_path, start_time, end_time  
FROM downloads;
```

Firefox:

places.sqlite + downloads.sqlite

C. Saved Passwords

Files:

- Chrome/Edge/Brave → Login Data
- Firefox → logins.json

These are encrypted using Windows DPAPI.

You need:

- Windows login password or NTLM hash
OR
- Forensic tools that bypass it (Axiom, Elcomsoft).

D. Cookies & Sessions

Chrome:

Cookies

Firefox:

cookies.sqlite

Extracted to view:

- authentication tokens
- session IDs
- tracking info

Tools: Nirsoft Cookie viewer / Autopsy

E. Cache Analysis

Stored in:

Cache\index, entries, .bin files

Tools:

- ChromeCacheView
- EdgeCacheView
- BrowserCacheViewer

Used to identify:

- images loaded from websites
- browsing activity
- IP addresses

F. Autofill Data

Chrome:

Web Data

Tables:

- autofill
- autofill_profiles

Check:

```
SELECT * FROM autofill;
```

Shows:

- Names
- Emails
- Phone numbers
- Addresses
- Sometimes card last 4 digits

G. Extensions (Malware/Proxy)

Chrome:

Extensions\

Each extension has folder names as unique IDs.

Check file:

manifest.json

Red flags:

- VPNs
- Proxy settings tools
- Screenshot/spy extensions
- Auto-forward extensions

Tools:

- ExtensionCrawler
- Autopsy → Browser Extensions

★ 4. QUICK PRACTICAL WORKFLOW YOU SHOULD FOLLOW

Here is a simple, real-world investigation flow:

Step 1: Locate browser folders

From HDD image → Extract:

- Chrome
- Edge
- Firefox
- Brave

Step 2: Load into Autopsy

Go to:

📁 Recent Activity Module

Autopsy will automatically generate:

- History timeline
- Download log
- Cookies list
- Login attempts
- User searches
- Sites visited
- Extensions installed
- Cache files

Step 3: For deeper analysis

Use NirSoft tools:

- BrowserHistoryView
- ChromeCacheView
- WebBrowserPassView
- ChromeCookiesView

Step 4: Check for Suspicious Activity

Look for:

- Logins to personal email / social media
- VPN websites
- Cloud storage (Drive, Dropbox)
- Dark web (.onion) access
- Malware download sites
- Proxy extensions

✓ 1. CHROME / EDGE / BRAVE — EXTENSION FORENSICS

All Chromium-based browsers store extensions in a similar structure.

📌 **Path:**

C:\Users\<USER>\AppData\Local\Google\Chrome\User Data\Default\Extensions\
For Edge:

C:\Users\<USER>\AppData\Local\Microsoft\Edge\User Data\Default\Extensions\
For Brave:

C:\Users\<USER>\AppData\Local\BraveSoftware\Brave-Browser\User Data\Default\Extensions\
Inside the **Extensions** folder you will see multiple folders like:

aapbdbdomjkkjkaonfhkkikfgjllcleb
cfhdojbkjhnklbpkdaibdccddilifddb
mhjfbmdgcfjbbpaeojofohoefgiehjai
These are extension IDs.

🔍 How to analyze an extension manually

1 Open the extension folder

Example:

Extensions\cfhdojbkjhnklbpkdaibdccddilifddb\1.1.8_0\

Every extension contains:

- manifest.json → **main configuration**
- .js files → **scripts (can be malicious)**
- Images/icons
- Background scripts

- Content scripts

🔥 **Very Important: Read manifest.json**

This file tells EVERYTHING about what the extension can do.

Open:

manifest.json

Look for:

✓ **1. Permissions**

If you see these permissions, suspect malicious intent:

- "proxy"
- "webRequest"
- "webRequestBlocking"
- "tabs"
- "history"
- "cookies"
- "management"
- "background"
- "clipboardWrite"
- "downloads"
- "nativeMessaging"

● **Red flags:**

- "https://*/" → full web access
- "*/**/*" → access to all URLs
- "storage" (normal but check what it stores)

✓ **2. Background scripts**

Example:

```
"background": {  
  "scripts": ["bg.js"]  
}
```

Open the script and check:

- API calls
- External communication
- Hardcoded URLs
- Cryptomining scripts
- Password stealing scripts

✓ **3. Content scripts**

Content scripts run on visited pages.

Example:

```
"content_scripts": [{  
  "matches": ["<all_urls>"],  
  "js": ["content.js"]  
}]
```

● If matches = <all_urls> → HIGH RISK

✓ 4. External connections

Look for:

"externally_connectable"

"content_security_policy"

If the extension calls external servers:

<http://123.45.67.89/api>

<https://malicious-domain.com/loader.js>

= CLEAR MALWARE.

🕵️ 2. Analyze extension folder contents

Check for:

- Minified JS (one-line JS files hiding malware)
- Obfuscated JS (eval(), base64 encoding)
- WebSockets / API calls
- *.exe downloads inside

⚡ QUICK TIP

Search within extension folder for keywords:

password

email

token

proxy

upload

cookie

send

ws://

http://

3. Tools to Analyze Extensions Automatically

✓ Autopsy

- Open Case
- Go to “Interesting Items” → “Chrome Extensions”
- Shows:
 - Extension name
 - Permissions
 - Description
 - Install time
 - Source URL

✓ Magnet AXIOM

- Amazing for extension forensics
- Shows:
 - Dangerous permissions
 - API calls
 - Communication domains
 - Scripts inside the extension
 - Timeline of extension activity

✓ NirSoft Extension Tools

ChromeExtensionsView

Shows:

- Enabled / Disabled
- Version
- Install Time
- Extension ID
- Permissions

✓ ExtensionCrawler

(Used in DFIR labs)

- Scans extension folder
- Flags sensitive permissions
- Identifies malicious behavior patterns

4. How to Identify Malicious Extensions Quickly

Look for:

1. Proxy-changing extensions

Permissions:

"proxy"

"webRequestBlocking"

● 2. Session hijacking extensions

Permissions:

"cookies"

"storage"

"tabs"

● 3. Keylogger / Spy extensions

Permissions:

"<all_urls>"

"activeTab"

"clipboardWrite"

● 4. Data exfiltration

Check JS files for:

fetch("http://...")

XMLHttpRequest(...)

● 5. Crypto-mining

Look for CPU-intensive JS, WebAssembly files.

📌 5. Firefox Extension Forensics

Firefox path:

C:\Users\<USER>\AppData\Roaming\Mozilla\Firefox\Profiles\<profile>\

File:

extensions.json

Contains:

- Name
- Permissions
- Version
- Install time
- Source

Extensions are stored here:

extensions\

★ SIMPLE PRACTICAL WORKFLOW

Step 1 — Extract extension folder

From HDD → path above

Step 2 — Open manifest.json

Check permissions + scripts.

Step 3 — Scan JavaScript files

Search keywords:

http, send, cookie, password, storage, websocket

Step 4 — Load image into Autopsy

Check extension metadata.

Step 5 — Use ExtensionCrawler or AXIOM

To identify malicious behavior.